bomb—which rests on the scale—will push harder on the scale than when the device is at rest in the same way that your body is squeezed back into the seat of an accelerating car. The bomb will ’squeeze’ the scale just as your back squeezes the cushion in the car seat. When a scale is squeezed, of course, its reading increases—and this will cause the bomb to detonate if the resulting increase is more than 50 percent.”

You thank Albert for his comment but, having tuned out his explanation to mentally confirm Isaac’s remark, you dejectedly proclaim that it takes only one fatal blow to kill an idea, and Isaac’s obviously correct observation has definitively done that. Feeling somewhat hopeless you ask for new suggestions. At that moment, Albert has a stunning revelation: “On second thought,” he continues, “I do not think that your idea is dead at all. Isaac’s observation that gravity diminishes as the device is lifted into space means that the reading on the scale will go down. My observation that the upward acceleration of the rocket will cause the device to push harder against the scale means that the reading will go up. Taken together, this means that if we carefully adjust the precise moment-to-moment acceleration of the rocket as it moves upward, these two effects can cancel each other out! Specifically, in the early stages of liftoff, when the rocket still feels the full force of the earth’s gravity, it can accelerate, just not too severely, so that we stay within the 50 percent window. As the rocket gets farther and farther from the earth—and therefore feels the earth’s gravity less and less—we need to increase its upward acceleration to compensate. The increase in the reading from upward acceleration can exactly equal the decrease in the reading from the diminishing gravitational attraction, so, in fact, we can keep the actual reading on the scale from changing at all!”

Albert’s suggestion slowly begins to make sense. “In other words,” you respond, “an upward acceleration can provide a stand-in or a substitute for gravity. We can imitate the effect of gravity through suitably accelerated motion.”

“Exactly,” responds Albert.

“So,” you continue, “we can launch the bomb into space and by judiciously adjusting the acceleration of the rocket we can ensure that the reading on the scale does not change, thus avoiding detonation until it is a safe distance from earth.” And so by playing off gravity